

Over-sampled Data Converters – ET4278

Practical Session

May 10th, 2019

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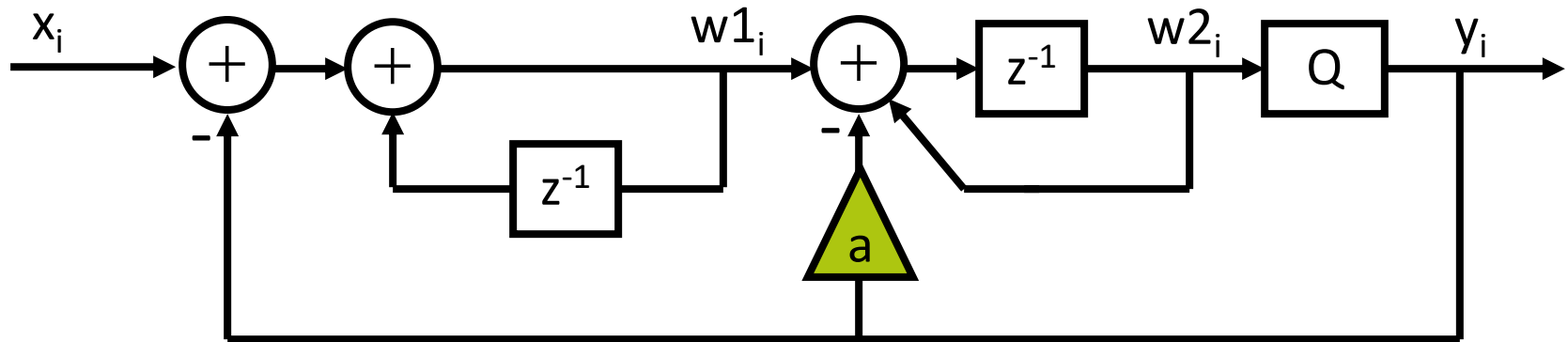
Today Overview

- 2nd order modulators
 - Feedback topology
 - Feed-forward topology
 - Local feedback
- Noise
- Continuous-Time (CT) modulators

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- **2nd order modulators**
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MOD2: Time Domain Simulation



Difference equations:

$$w2_i = w2_{i-1} + w1_{i-1} - \mathbf{a}y_{i-1}$$

$$y_i = Q(w2_i)$$

$$w1_i = w1_{i-1} + x_i - y_i$$

Exercise 2.1

- Download (from Blackboard) and run the m-file: MOD2.m
 - This m-file simulates the behaviour of a 2nd order modulator whose input is a DC offset $\mathbf{x} = 0$
 - Run the code and examine the accumulator outputs **w1** and **w2**, as well as the bitstream **y**
- Apply DC offsets of 50 mV and 34.6 mV to the modulator.
 - Observe the changes in **w1**, **w2** and **y**.
 - Compute the bitstream average and calculate the quantization error. Experiment with different values of $\mathbf{x}$ and with sinc¹ and sinc² decimation filters. Is there a difference in the Q-error?
 - Change the “nr_periods” from 100 to 1000. By what factor does the quantization error change? Can you explain this?
- Observe the stabilizing coefficient “a” in the code.
 - For the same input value of 34.6mV start reducing “a”, while studying the waveforms and the quantization error for both filtered and un-filtered outputs.
 - What happens to the modulator when $a = 0$? Why?

Exercise 2.1: Discussion

- For DC values that are simple integer ratios, MOD2 still exhibits periodic limit cycles, but they are more chaotic than those of the 1st order modulator
- For a given “nr_periods” MOD2 has (in general) less Q-error than MOD1. Furthermore, this error decreases more rapidly as nr_periods is increased.
- Use of a sinc² filter results in significantly less Q-error
- According to the linear model, if nr_periods is doubled, by what factor should the Q-error decrease?

Exercise 2.1: Discussion Cont'd

- According to the linear model the in-band noise power of a 2nd order modulator is given by:

$$n_{rms}^2 = \frac{\pi^4 q_{rms}^2}{5 \cdot OSR^5}$$

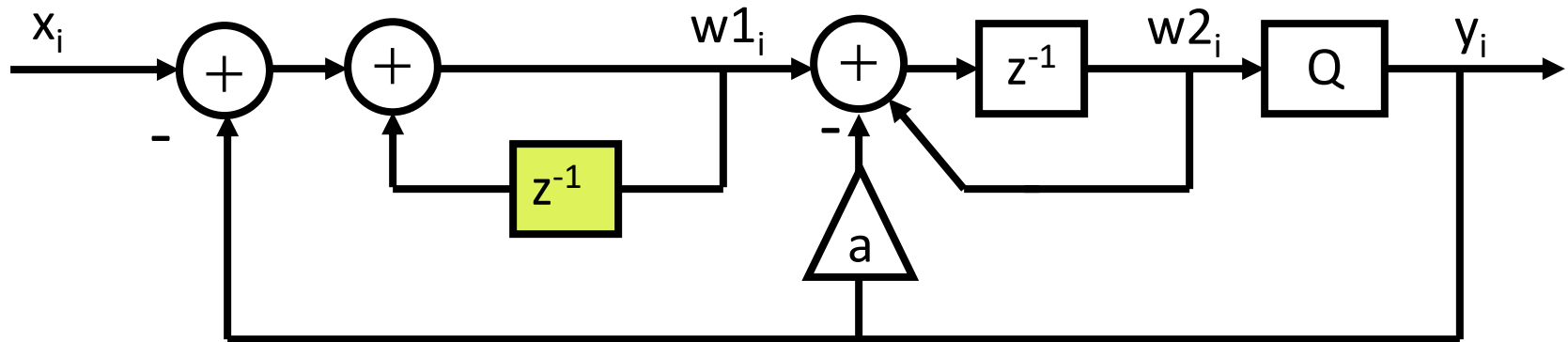
- So if the OSR doubles, the noise level should decrease by about $4\sqrt{2}$ i.e. a factor of 5.7 or 15 dB

$$\begin{aligned}\Delta SNR &= 10 \log \Delta OSR^5 \\ &= 2.5(20 \log \Delta OSR)\end{aligned}$$

Exercise 2.1: Discussion Cont'd

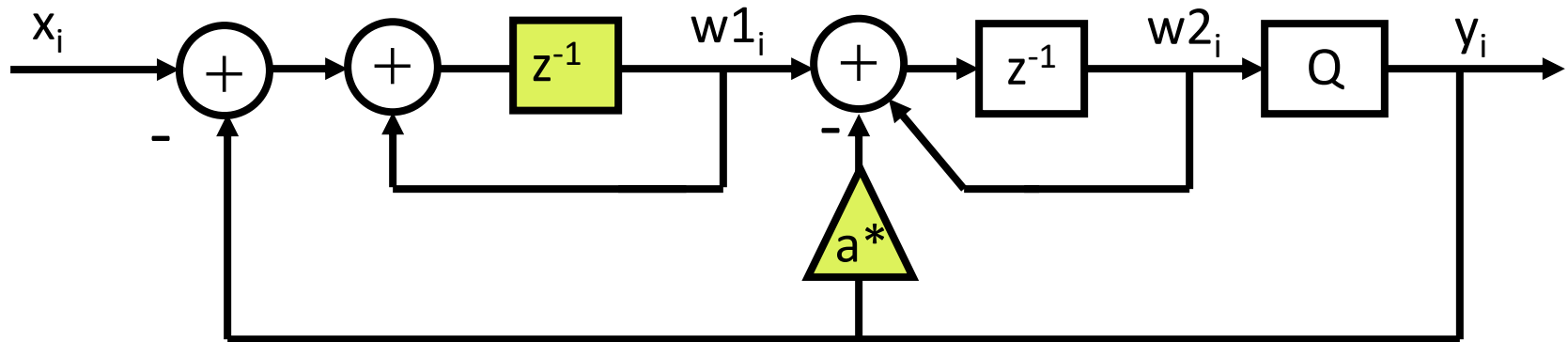
- What was the difference between the original FIR filter and a sinc^2 filter? Can you explain this?
- As “a” **decreases**, the amplitude of w_1 and w_2 **increases** until the modulator becomes **unstable** for $a = 0$
- Since a SDM is a clocked oscillator, the term stability means something different than it does for linear systems!

MOD2: Delayed vs. Delay-free Accumulator



- Until now, you have simulated with a **delay-free** 1st accumulator

MOD2: Delayed vs. Delay-free Accumulator



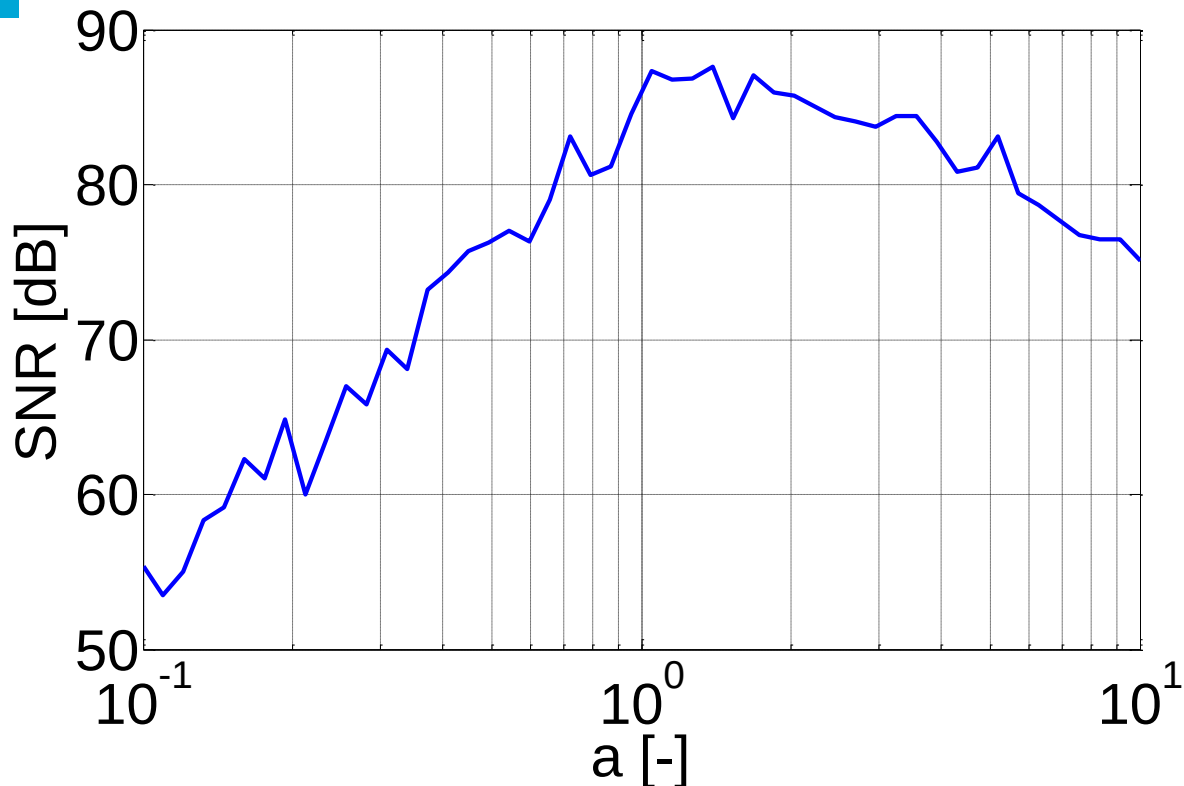
- This is the same architecture used in theory class
- Which stabilizing coefficient should be used? $a^* = a$? What is the effect on NTF and STF?

Exercise 2.2

- Download the file: MOD2_SNR.m
 - This code applies a single tone sinusoid signal to the input of the 2nd order sigma delta modulator.
 - Run the code and study the bitstream and the shape of the quantization noise at the output spectrum.
 - Pay attention to the 3rd plot defining the band of interest, where SNR is calculated.
- Start reducing the stabilizing coefficient “a”, e.g. $a = 0.1$
 - Study its effect on the spectrum
 - What happens to the SNR as you reduce “a”? Why?
- Modify the script to implement a delayed 1st accumulator.
 - Find the new value for a^* .
 - Do you observe differences in spectrum or SNR?

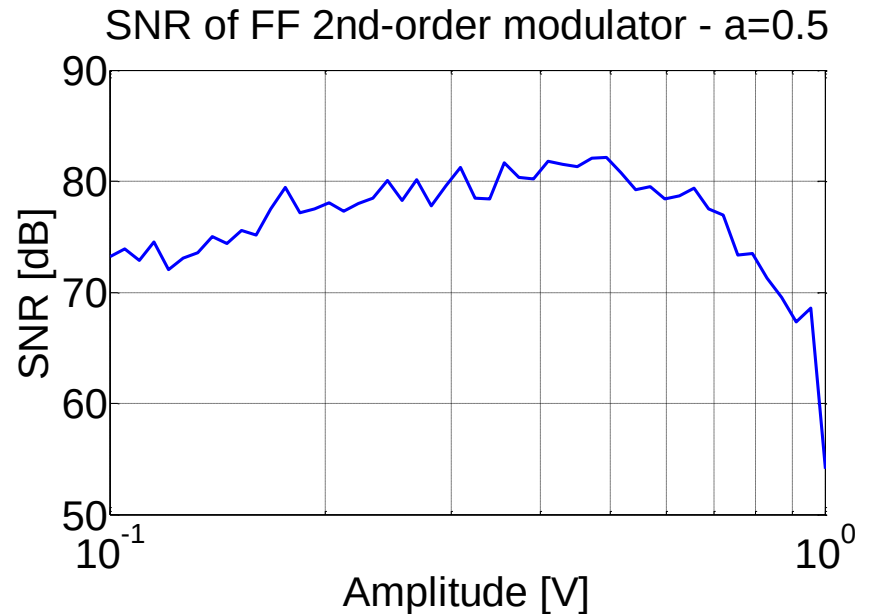
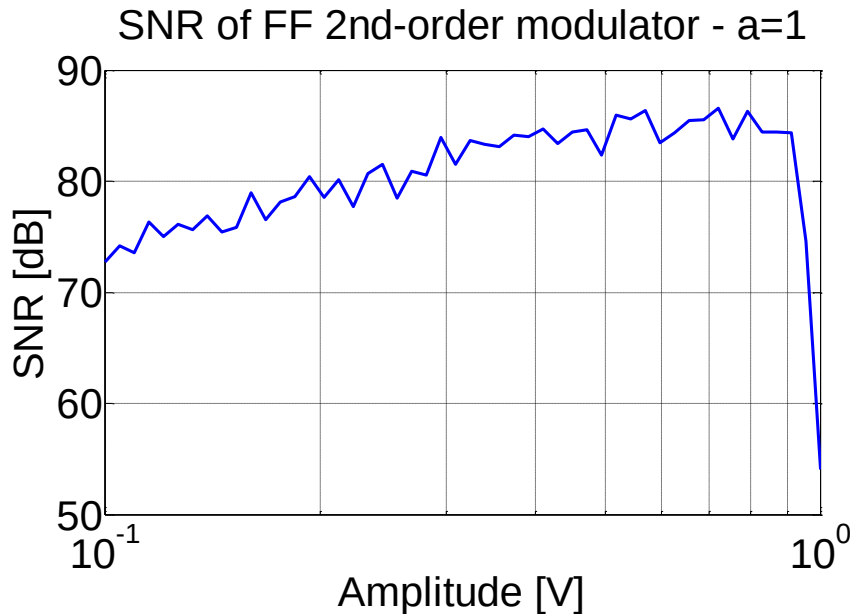
Exercise 2.2: Discussion

SNR of FF 2nd-order modulator



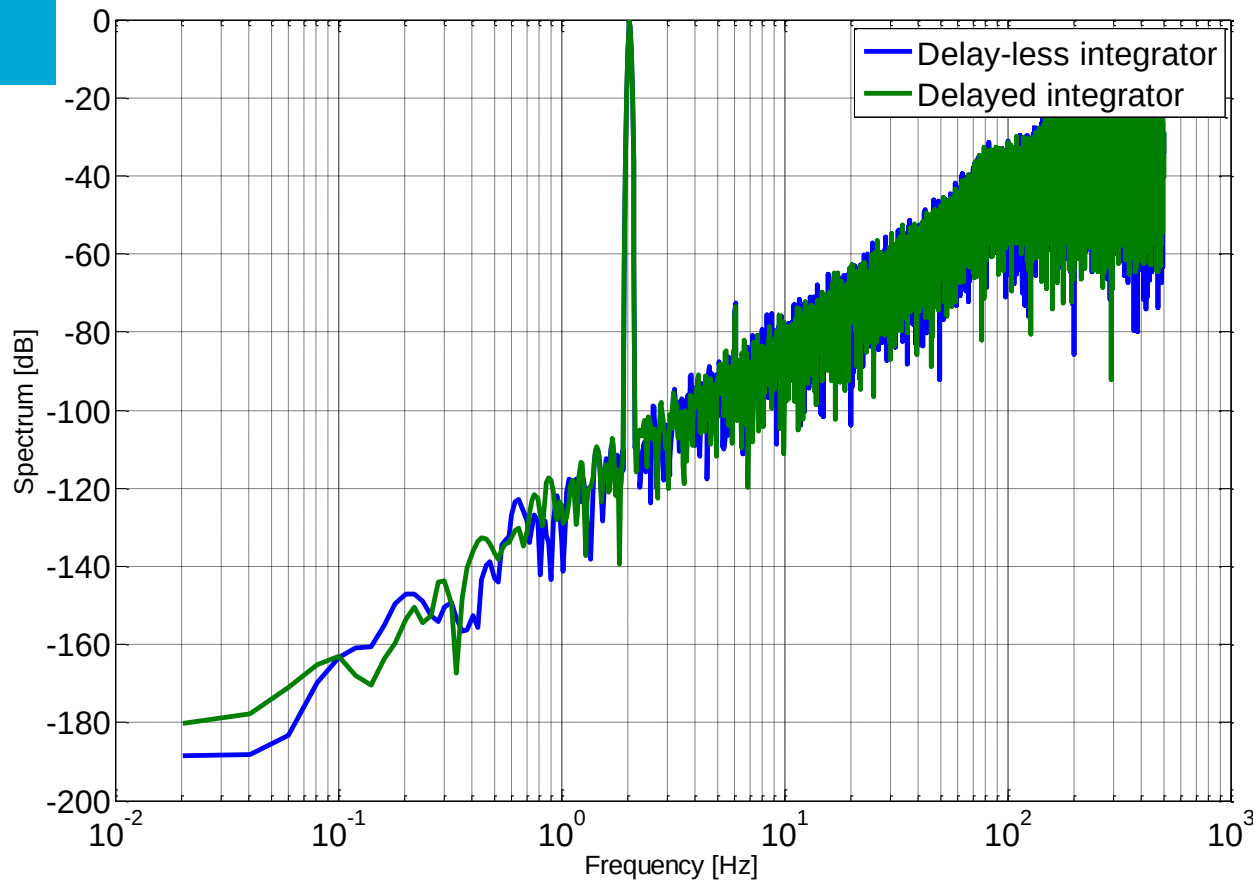
- SNR has a maximum for $a \sim 1$
- But this was obtained for $A = 0.7$
- Is this optimal?

MOD2: SNR vs Amplitude



- For $a = 1$, A can be almost equal to 1
- $f_{in} = 2$, $f_b = 4$, $f_s = 1000$

MOD2: Delayed vs. Delay-free Accumulator

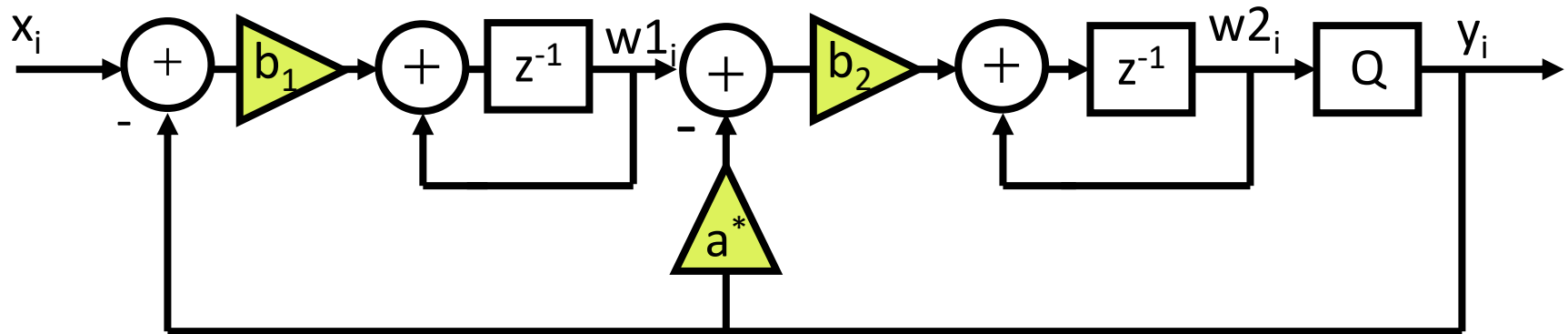


- Linear model
 - Same loop filter $H(z)$
 - $a = 1$
 - $a^* = 2$
- Simulations
 - $\text{SNR}_{\text{delayed}} = 86 \text{ dB}$
 - $\text{SNR}_{\text{delayless}} = 84 \text{ dB}$

Modulator Scaling

- In a practical implementation, it is important to ensure that the accumulators' output signals lie within bounds determined by the supply rails and by amplifier topology.
- This is done by adjusting the coefficients of the loop filter. This process is known as **modulator scaling**.
- In the case of a 1-bit quantizer, the amplitude of the internal signals can be easily adapted without altering the NTF or STF.
- For multi-bit quantizers , this is not so easy and the amplitude of the reference signals may also need to be scaled.

Scaled FB Modulator (1)



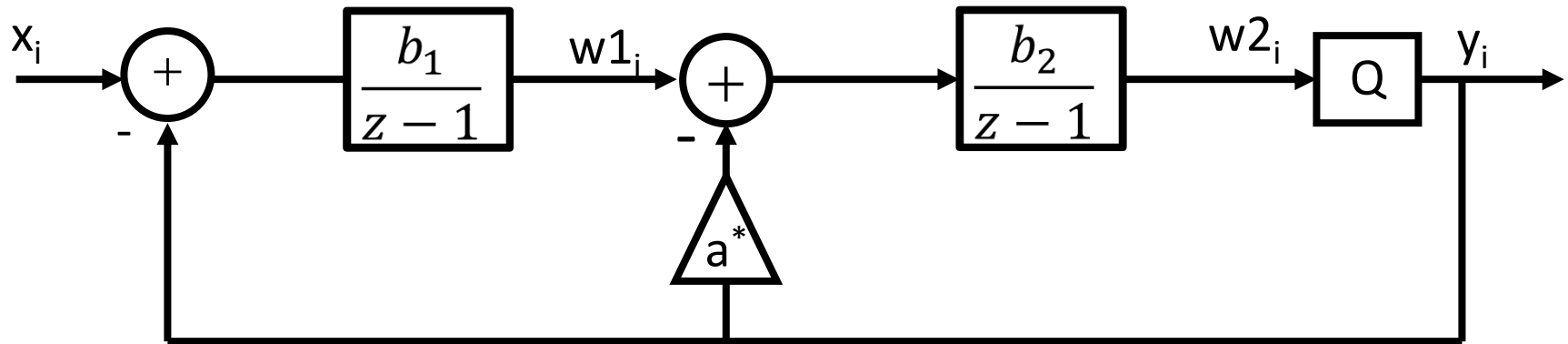
Scaled Difference equations

$$w2_i = w2_{i-1} + b_2(w1_{i-1} - \mathbf{a}^*y_{i-1})$$

$$y_i = Q(w2_i)$$

$$w1_i = w1_{i-1} + b_1(x_i - y_i)$$

Scaled FB Modulator (2)



Before scaling:
$$H(z) = \frac{az + 1 - a}{(z - 1)^2}$$

After scaling:
$$H^*(z) = b_2 \frac{a^*z + b_1 - a^*}{(z - 1)^2} = b_1 b_2 H(z) \text{ if } a^* = ab_1$$

Exercise 2.3

- Download (from Blackboard) and run the m-file: MOD2_FB.m
 - This m-file simulates a 2nd order **feed-back** modulator with a sinusoidal input signal.
 - Run the code and examine the time & frequency domain graphs.
- Scale the accumulator gain coefficients
 - For an input amplitude of 0.7 V, adjust the gain coefficients so that the output swing of both accumulators is less than 0.2 V.
 - Ensure that your scaling does not affect modulator stability!
 - What happens to the modulator's STF?

Answer to Exercise 2.3

- Initial coefficients:

b1 = 1; b2 = 1;

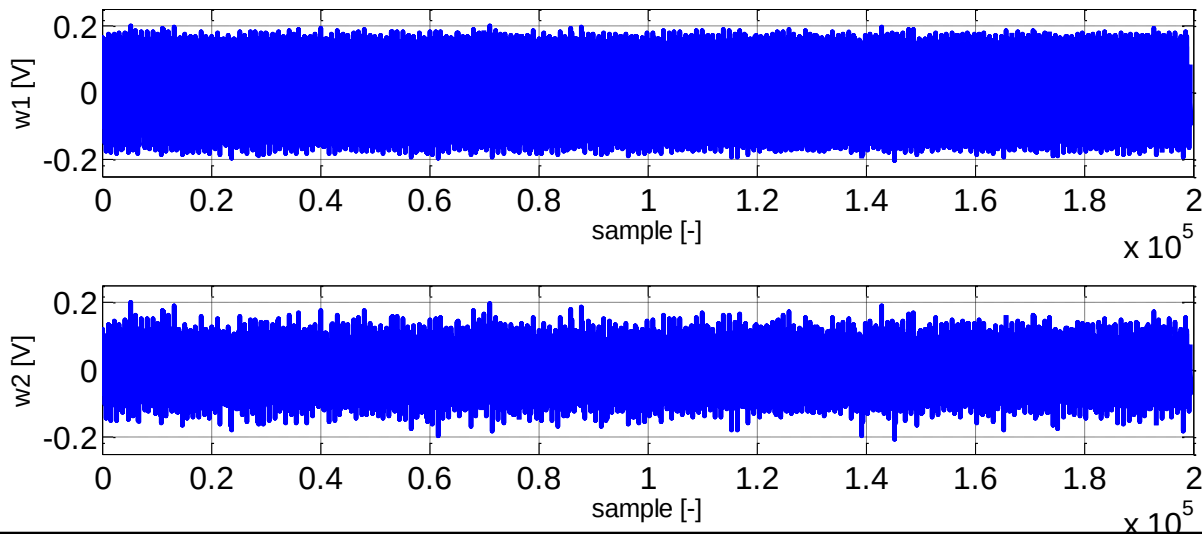
a1 = 1; a2 = 2;

- For accumulator output swing < 0.2 V, in Matlab after running the original modulator:

b1_scaled = b1*0.2/max(abs(w1)); % b1_scaled = 0.06

a2_scaled = a2*b1; % a2_scaled = 0.13

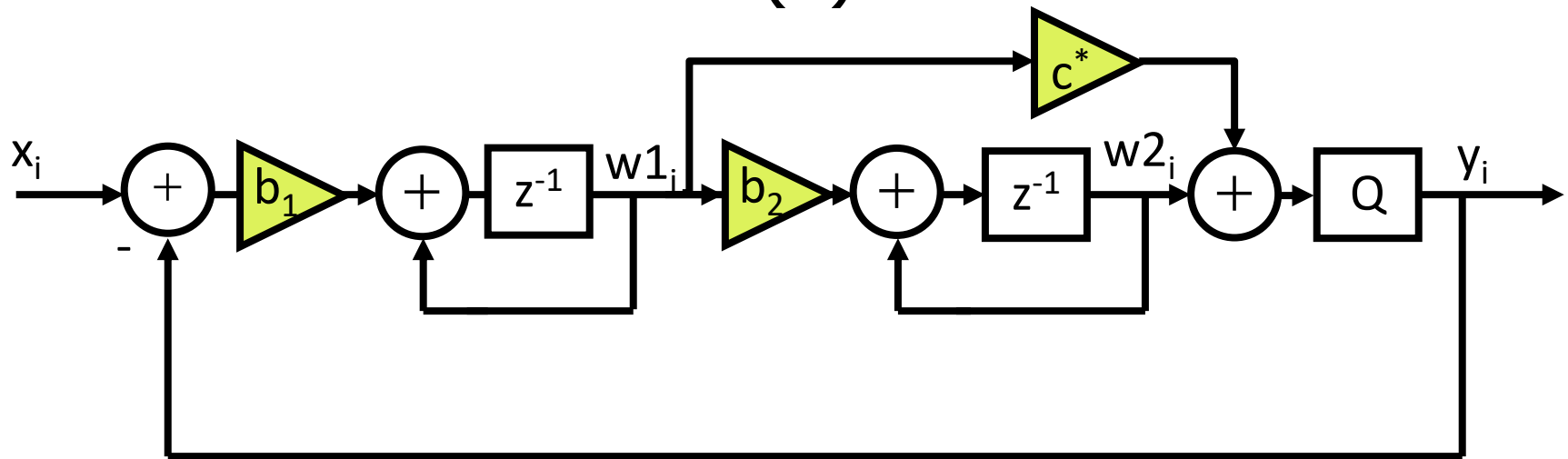
b2_scaled = b2*0.2/max(abs(w2))/b1; % b2_scaled = 0.5



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Scaled FF Modulator (1)



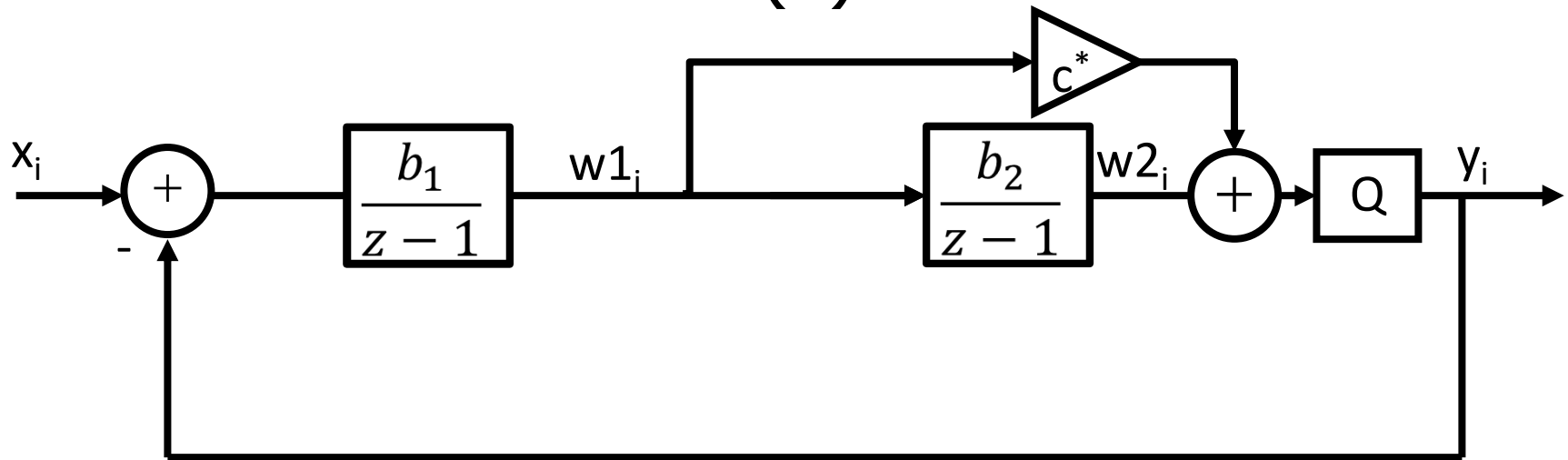
Add extra coefficients b_1 & $b_2 \Rightarrow$ Scaled difference equations:

$$w2_i = w2_{i-1} + b2 * w1_{i-1}$$

$$y_i = Q(w2_i + c * w1_i)$$

$$w1_i = w1_{i-1} + b_1(x_{i-1} - y_{i-1})$$

Scaled FF Modulator (3)



Before scaling:
$$H(z) = \frac{cz + 1 - c}{(z - 1)^2}$$

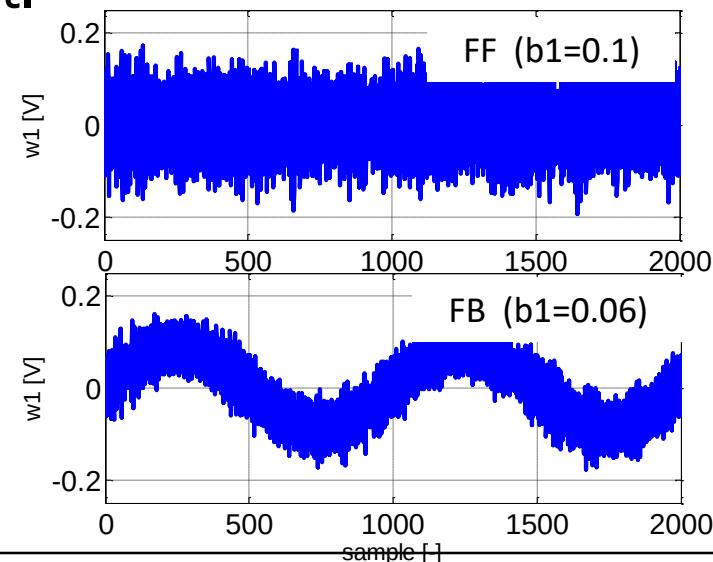
After scaling:
$$H^*(z) = b_1 \frac{c^*z + b_2 - c^*}{(z - 1)^2} = b_1 b_2 H(z) \text{ if } c^* = c b_2$$

Exercise 2.4

- Download (from Blackboard) and run the m-file: MOD2_FF.m
 - This m-file simulates a 2nd order **feed-forward** modulator with a sinusoidal input signal.
 - Run the code and observe the time and frequency domain graphs.
 - How does the 1st accumulator's output signal compare with that of a feed-back modulator? Why are they different?
- Scale the accumulator gain coefficients
 - For an input amplitude of 0.7, adjust the gain coefficients so that the output swing of both accumulators is less than 0.2 V.
 - Ensure that your scaling does not affect modulator stability!
- Modify the code to simulate a delay-less 2nd accumulator
 - What happens?
 - How to rescale the coefficient c^* for stability?

Answer to Exercise 2.4

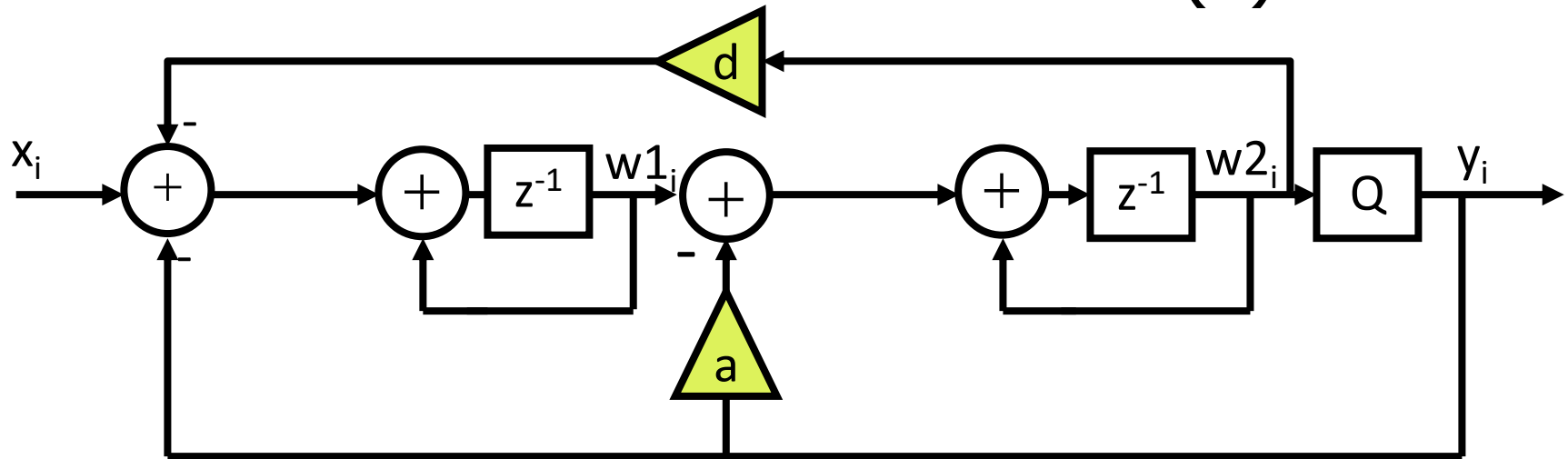
- In the feed-forward topology, the 1st accumulator's output signal is a high-pass filtered version of the input signal, and therefore only high-frequency quantization noise is present there
- In the feed-back topology, the 1st accumulator's output signal is a low-pass filtered version of the input signal, and so the low-frequency input signal will also be present.
- For swings less than 0.2 V:
 $b1 = 0.1$; $b2 = 0.3$;
 $c1 = \underline{2*b2} = 0.6$; $c2 = 1$;



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Scaled Modulator with Resonator (1)



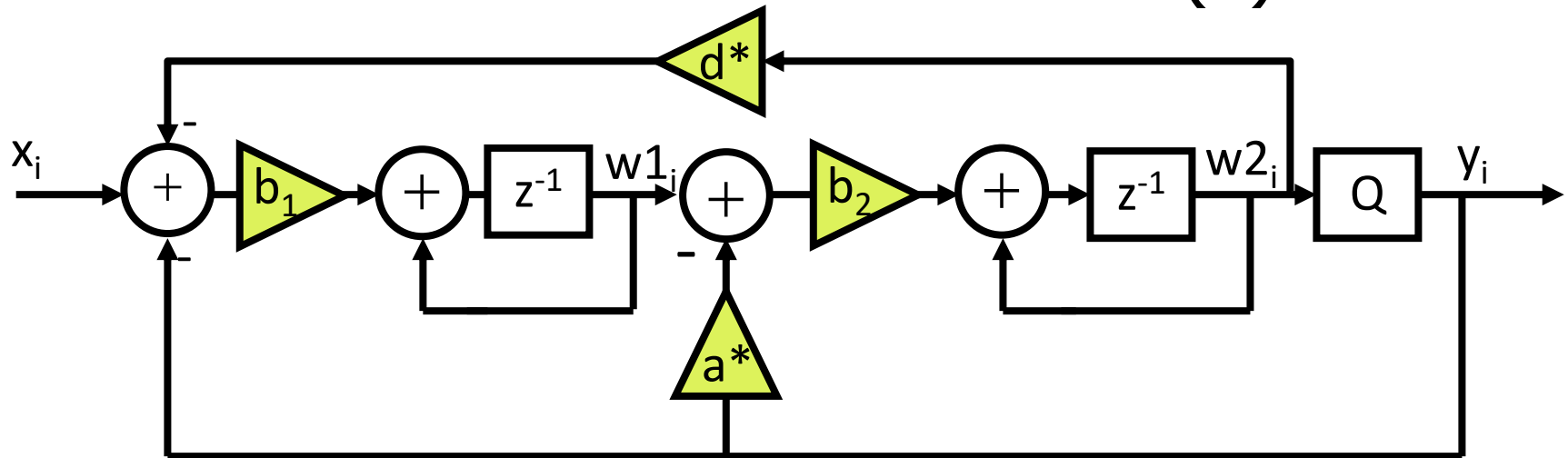
Difference equations

$$w2_i = w2_{i-1} + w1_{i-1} - a \cdot y_{i-1}$$

$$w1_i = w1_{i-1} + x_{i-1} - y_{i-1} - d \cdot w2_{i-1}$$

$$y_i = Q(w2_i)$$

Scaled Modulator with Resonator (2)



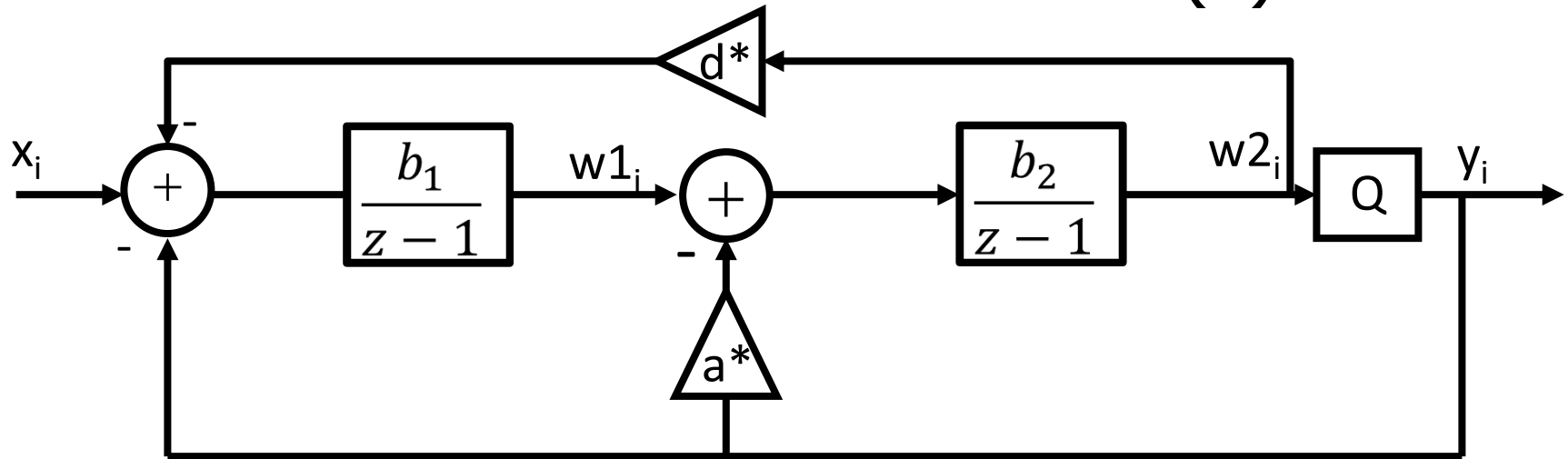
Difference equations

$$w2_i = w2_{i-1} + \mathbf{b}_2(w1_{i-1} - \mathbf{a}^* \cdot y_{i-1})$$

$$w1_i = w1_{i-1} + \mathbf{b}_1(x_{i-1} - y_{i-1} - \mathbf{d}^* \cdot w2_{i-1})$$

$$y_i = Q(w2_i)$$

Scaled Modulator with Resonator (3)



Before scaling:

$$H(z) = \frac{az + 1 - a}{(z - 1)^2 + d}$$

$$H^*(z) = b_2 \frac{a^*z + b_1 - a^*}{(z - 1)^2 + db_1b_2} = b_1b_2H(z) \text{ if } a^* = ab_1$$

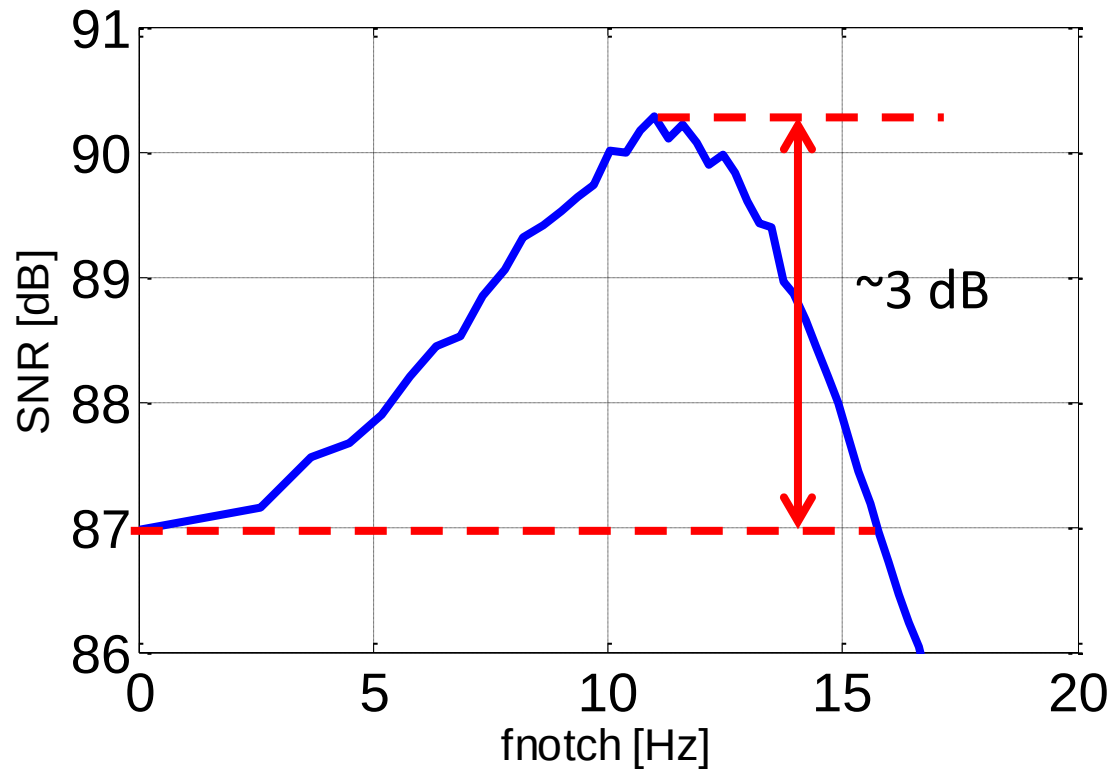
After scaling

$$\text{and } d^* = \frac{d}{b_1b_2}$$

Exercise 2.5

- Download (from Blackboard) and run the m-file: MOD2_FB_localFB.m
 - This m-file simulates a 2nd order feedback modulator with local feedback and sinusoidal input signal.
 - Initially, set $d = 0$.
 - Run the code and observe the time and frequency domain graphs.
- Choose a value for d
 - Observe the relation between notch frequency and d .
 - Choose d to maximize the SNR
- Scale the coefficients
 - For an input amplitude of 0.5, adjust the gain coefficients so that the output swing of both accumulators is less than 0.2 V.

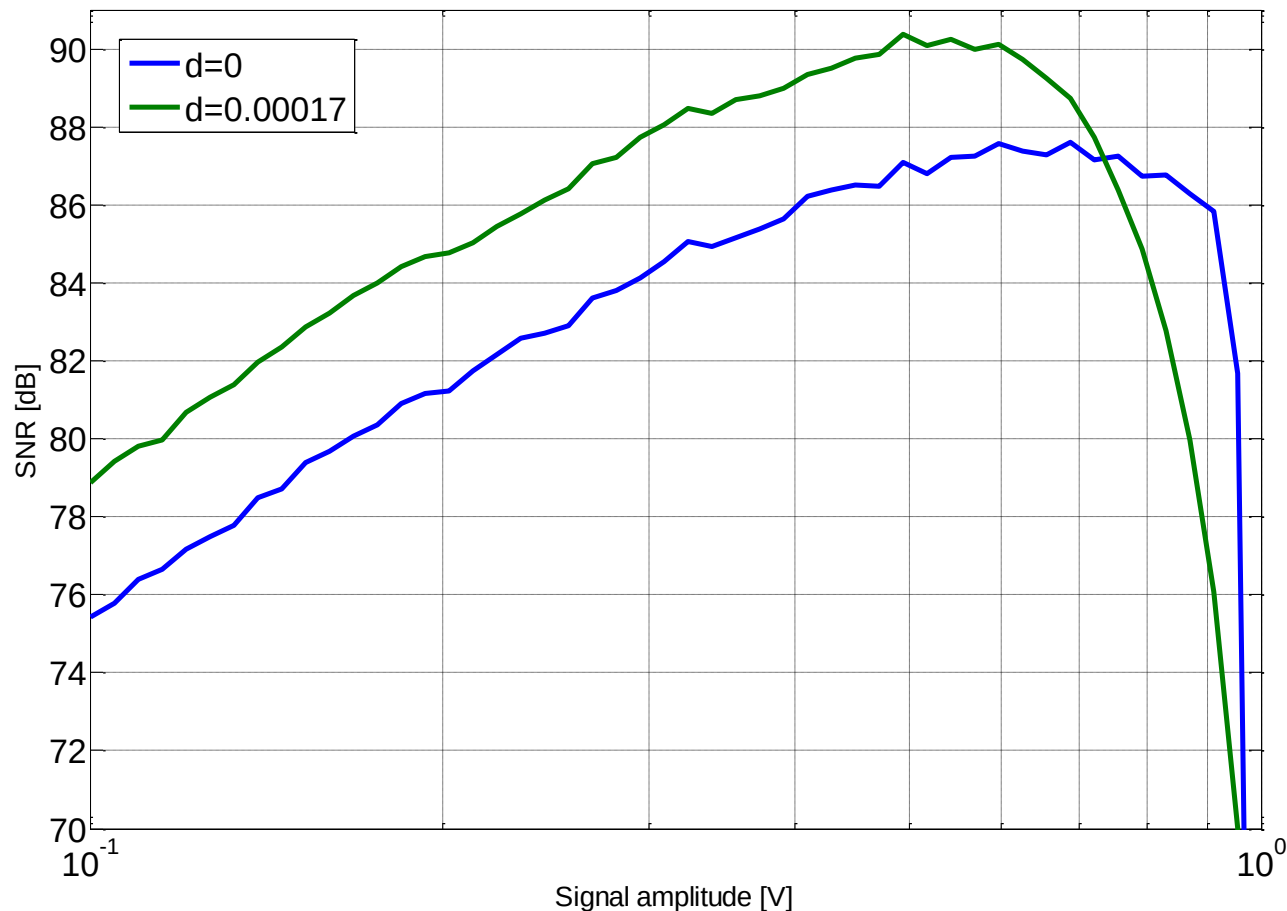
Answer to Exercise 2.5 (1)



$A = 0.5 \text{ V}$

- Optimal $f_{\text{notch}} = f_b / \sqrt{3} = 11.5 \text{ kHz}$ [Schreier]

Answer to Exercise 2.5 (2)



- But optimizing f_{notch} reduces the maximum input signal

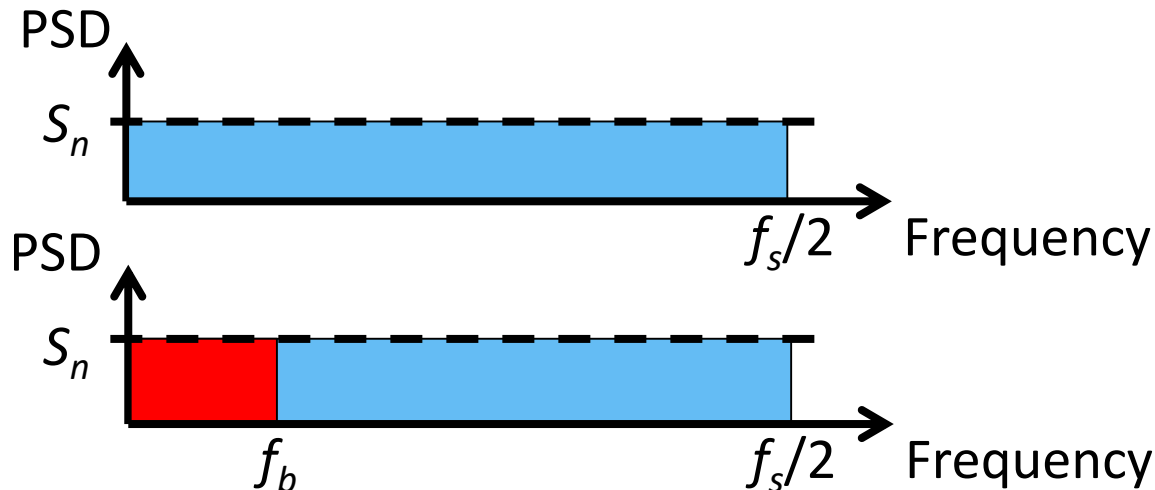
Answer to Exercise 2.5 (3)

- Before scaling:
 $b1 = 1; b2 = 1;$
 $a1 = 1; a2 = 2;$
 $d = 0.00017;$
- For swings less than 0.2 V:
 $b1 = 0.06; b2 = 0.5;$
 $a1 = 1; a2 = 2*b1 = 0.12;$
 $d = 0.00017/b1/b2 = 0.005;$

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Oversampling Thermal Noise



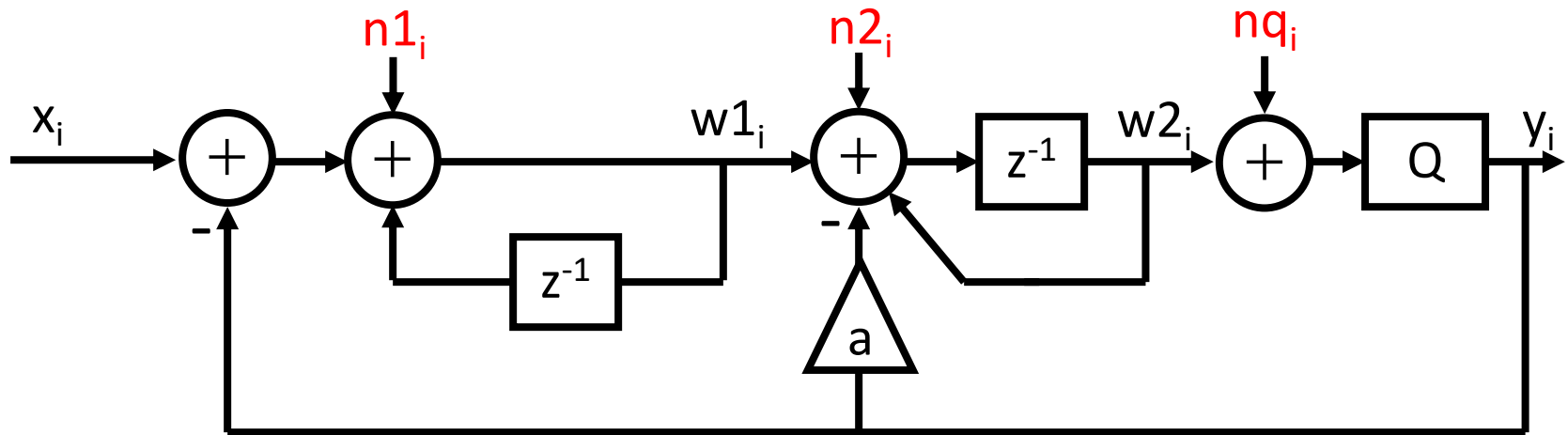
$$v_{total,rms} = S_n \cdot f_s/2$$

$$v_{in-band,rms} = S_n \cdot f_b$$
$$= \frac{v_{total,rms}}{OSR}$$

- Generating white noise in Matlab with RMS value equal to noise_rms

```
for i=1:N,  
    n(i) = noise_rms * randn; % white Gaussian noise  
end;
```

MOD2 with Thermal Noise



Difference equations

$$w2_i = w2_{i-1} + w1_{i-1} - ay_{i-1} + \mathbf{n2_{i-1}}$$

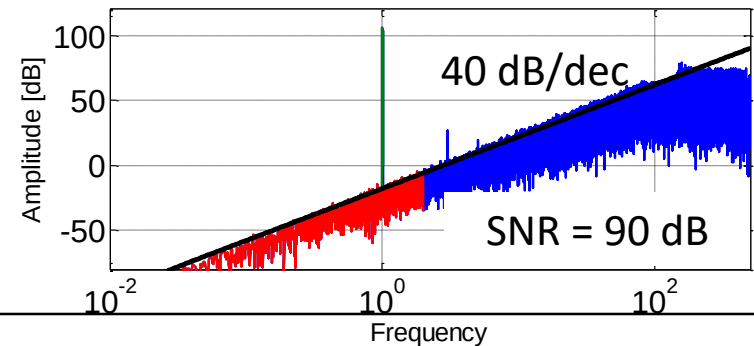
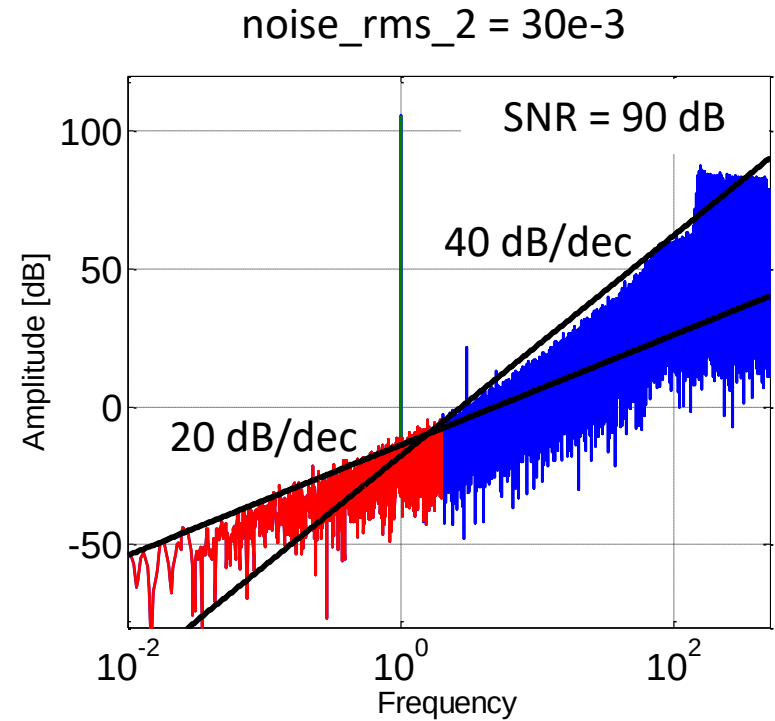
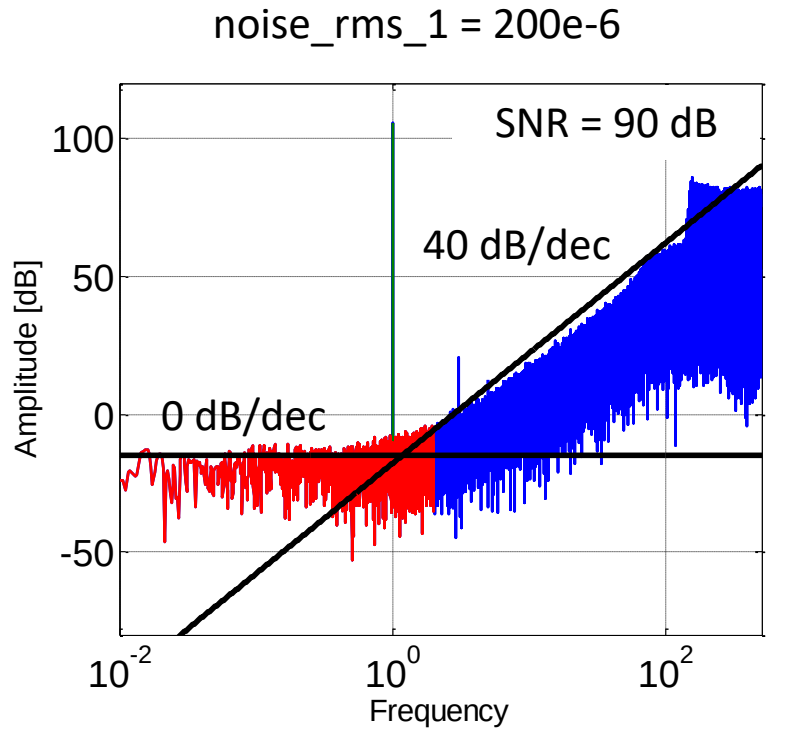
$$y_i = Q(w2_i + \mathbf{nq_i})$$

$$w1_i = w1_{i-1} + x_i - y_i + \mathbf{n1_i}$$

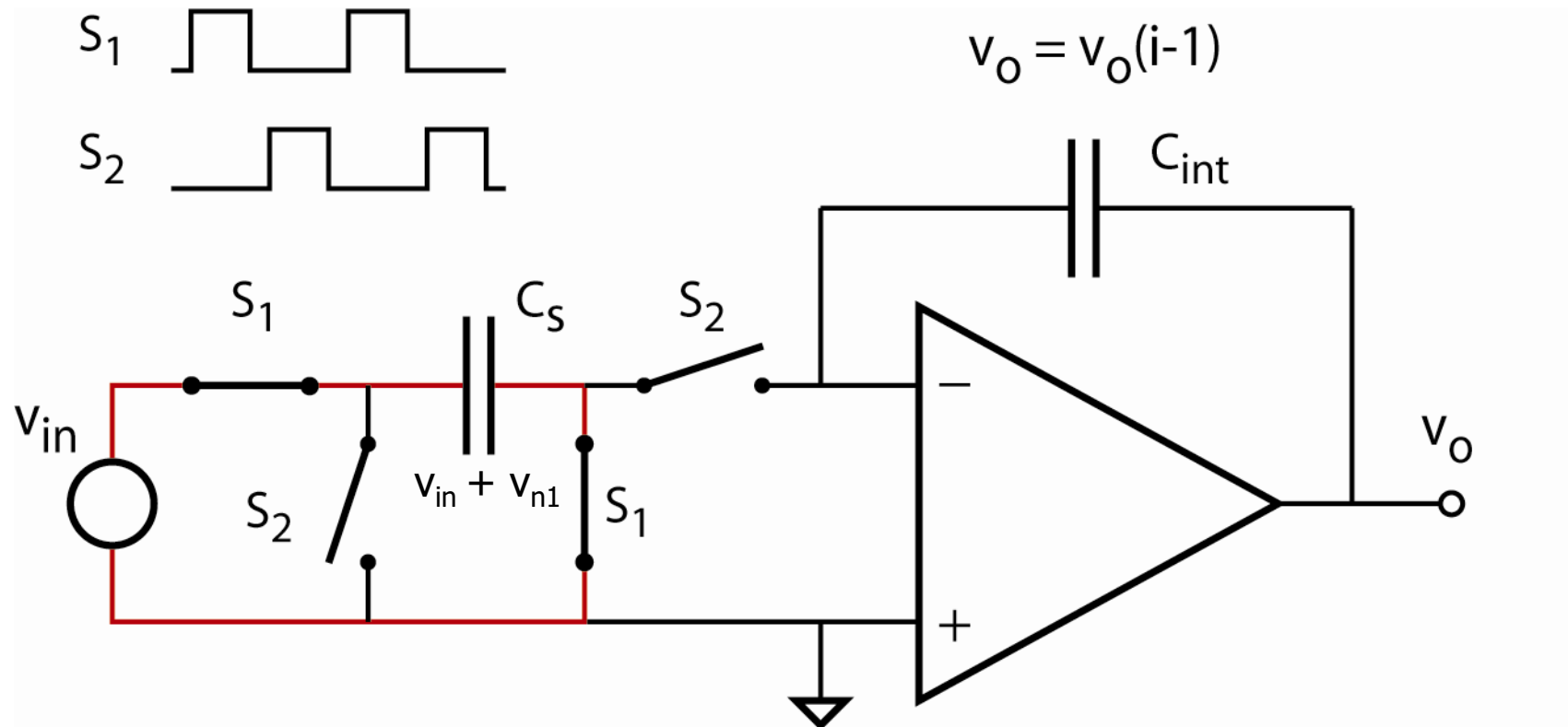
Exercise 2.6

- Download (from Blackboard) and run the m-file: MOD2_noise.m
 - This m-file simulates a 2nd order feedback modulator with noise.
 - Run the code for a DC input of 0 V and 34.6 mV, without noise.
 - Then repeat the simulations with 1st accumulator RMS noise of 200 μ V ($\text{noise_rms_1} = 100\text{e-}6$).
 - Observe the output spectrum. Do you see changes with and without noise? Why?
- Simulate a sinusoidal input with 0.7 V amplitude (DC offset = 0)
 - How much noise is needed at the 1st integrator input for SNR = 90 dB?
 - Modify the code to include noise **only** at the input of 2nd accumulator. How much noise is needed for SNR = 90 dB?
 - Modify the code to include noise **only** at the input of quantizer. How much noise is needed for SNR = 90 dB?

MOD2 with Thermal Noise



Switched Capacitor Accumulator (1)



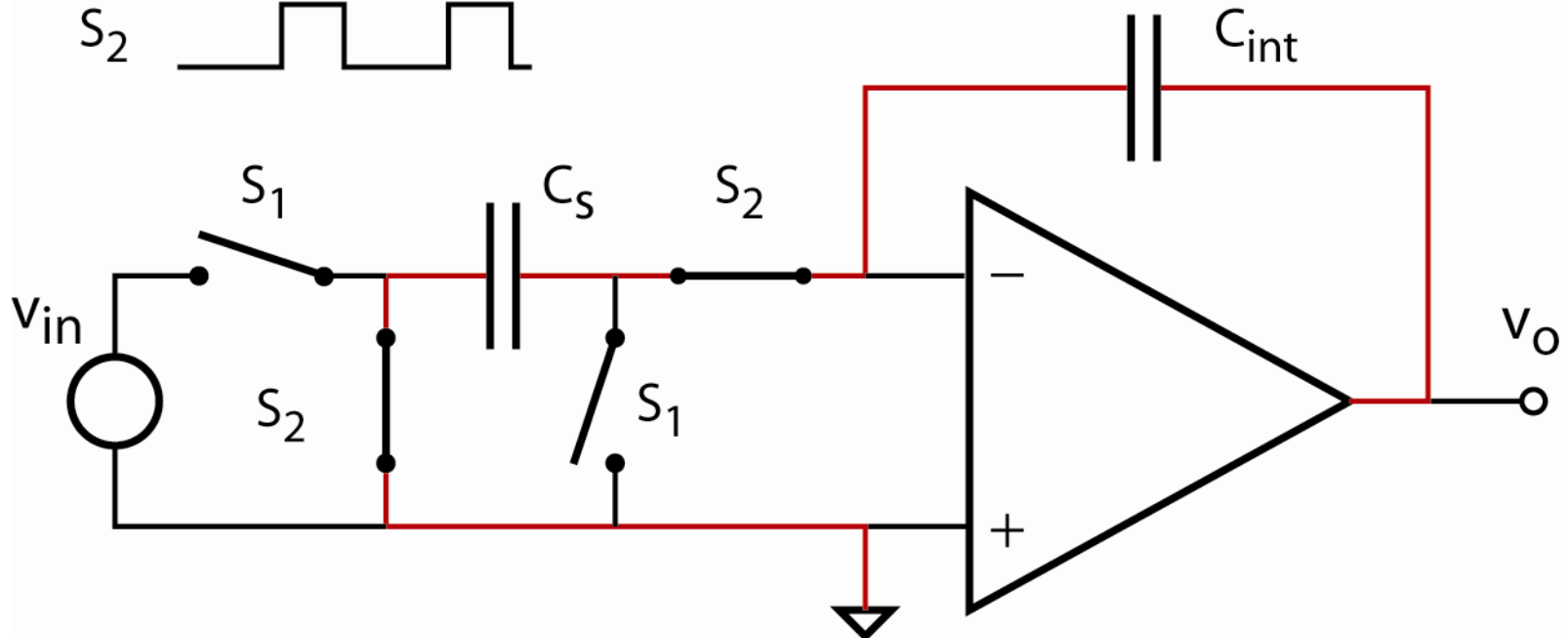
- Input voltage is sampled on C_s
- Noise is also sampled on C_s : $\langle v_{n1}^2 \rangle = kT/C_s$

Switched Capacitor Accumulator (2)

S_1

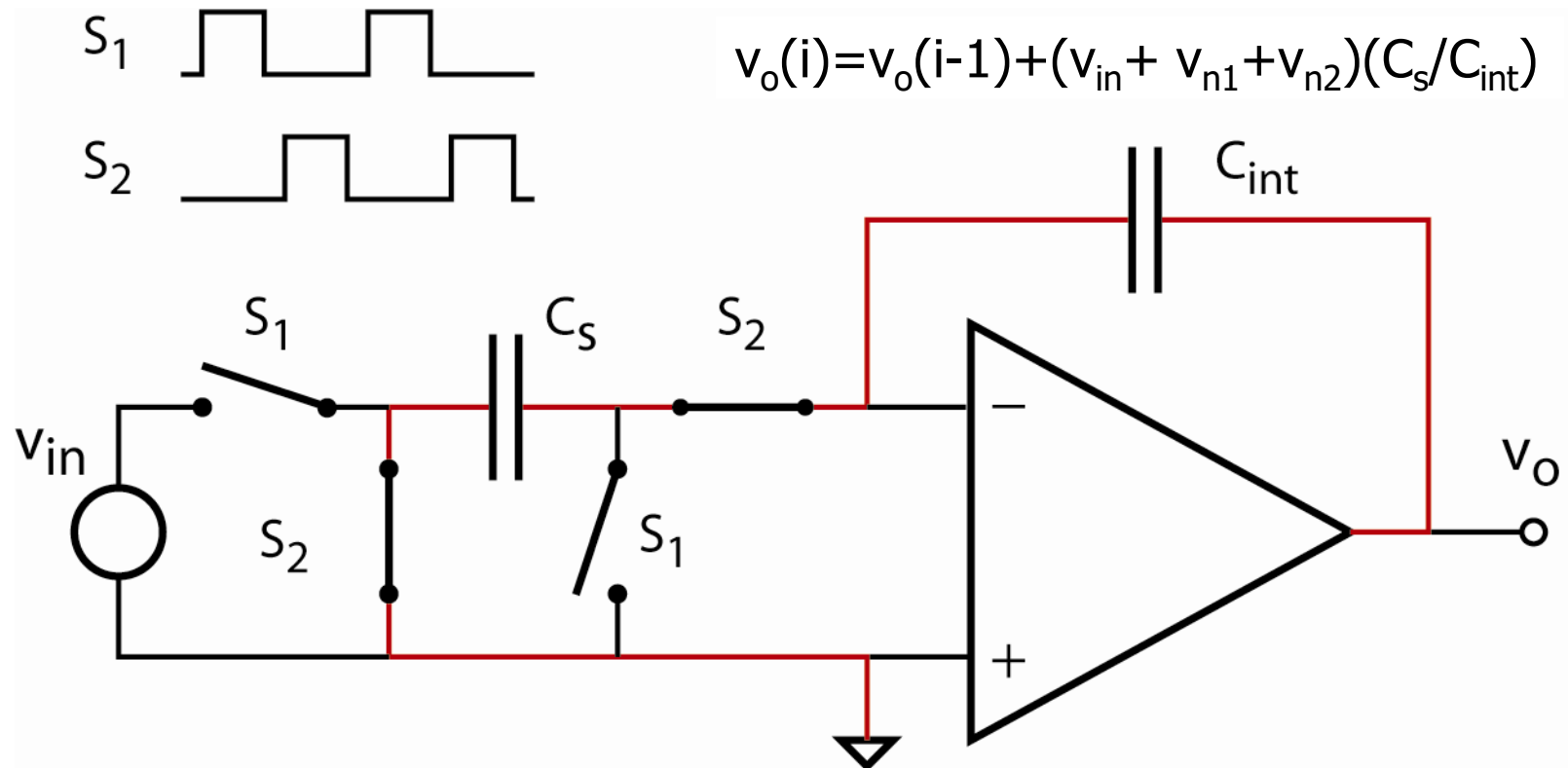
S_2

$$v_O(i) = v_O(i-1) + v_{in}(C_S/C_{int})$$



- Sampled voltage is transferred to C_{int}
- At opening of S_2 additional noise v_{n2} is transferred to C_{int} :
 $\langle v_{n2}^2 \rangle = kT/C_S$

Switched Capacitor Accumulator Noise



- Input referred noise: $\langle v_n^2 \rangle = \langle v_{n1}^2 + v_{n2}^2 \rangle = 2kT/C_s$
- Neglecting opamp noise.

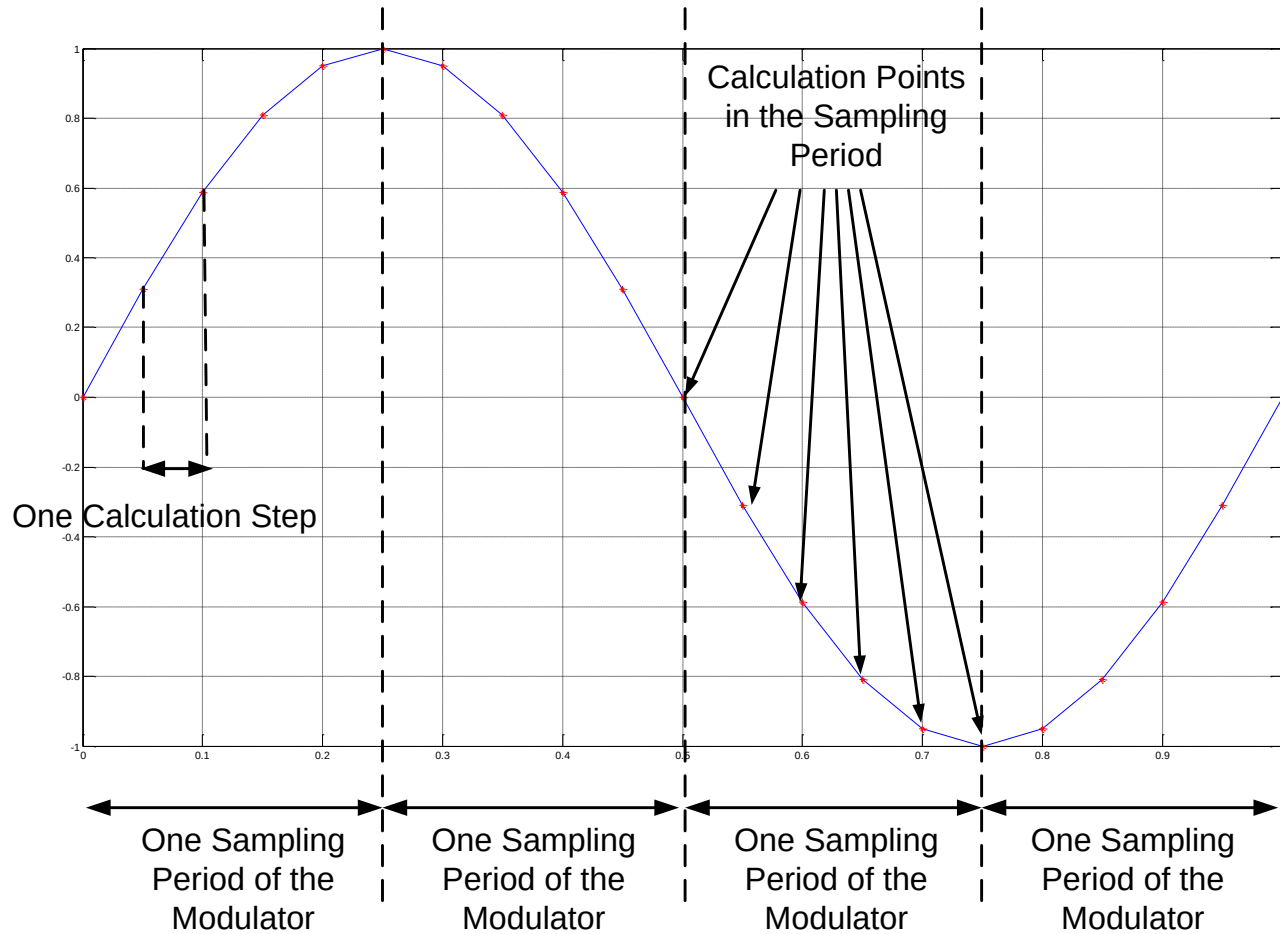
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- **Continuous-Time (CT) modulators**

Simulating Continuous Time (CT) Modulators

- **MATLAB** is a **discrete time** environment!
- But the output of the loop filter of a CT sigma-delta modulator is a **continuous time** signal.
- How can this be modeled in MATLAB?
- **Solution:** over-sample each sampling period!
 - Each sampling period is divided into a number of calculation steps.
 - After each step, the internal signals of the loop filter are evaluated.
 - At the end of every sampling period, the output of the loop filter is used to update the output of the quantizer.
 - The new quantizer output is then used to update the state of the loop filter during the next sampling period.

Simulating CT Sigma-Delta Modulators



Code for Simulating a CT Modulator (1)

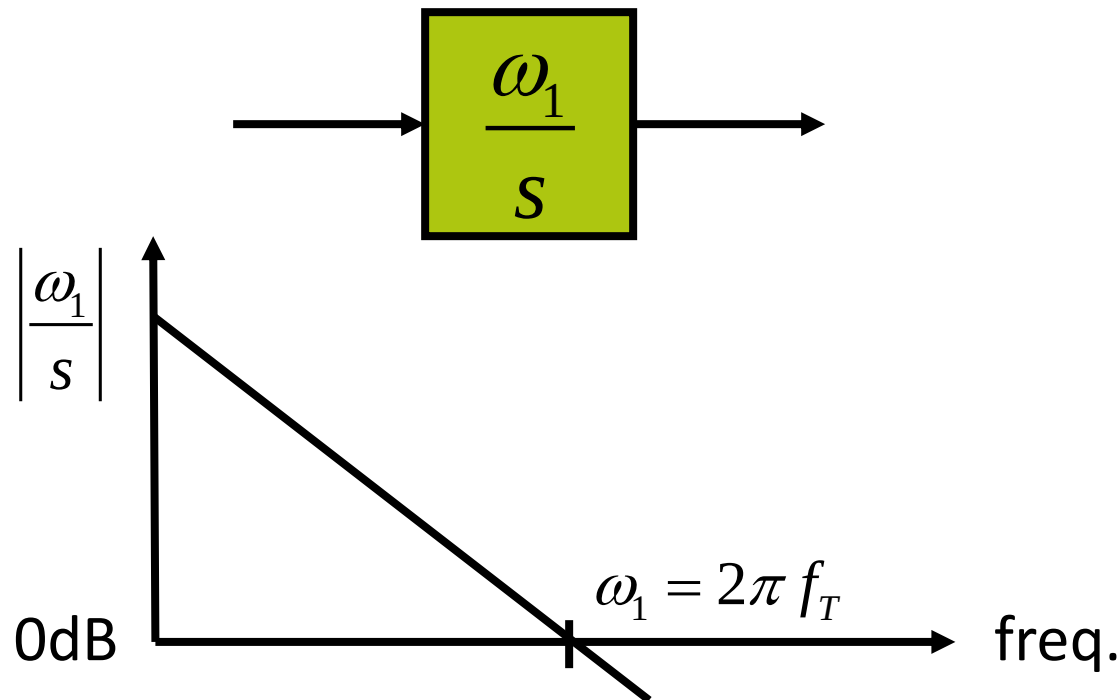
- **Two** nested “for” loops are required – a main loop and an internal loop.
- Each iteration of the main loop represents one sampling period.
- Each iteration of the internal loop represents one calculation step
- The quantizer is evaluated at the end of the internal loop.

```
ix = 1;
for a = 2:nr_points,
    for b = 1:nr_steps,
        ix = ix+1;
        w2(ix) = w2(ix-1) + omega2*[w1(ix-1) - a2*y(a-1)];
        w1(ix) = w1(ix-1) + omega1*[in(ix-1) - a1*y(a-1)];
    end;
    y(a) = 2*(w2(ix)>=0)-1;
End;
```

- “nr_points” is the number of samples
- “nr_steps” is the number of calculation steps **per sampling period**

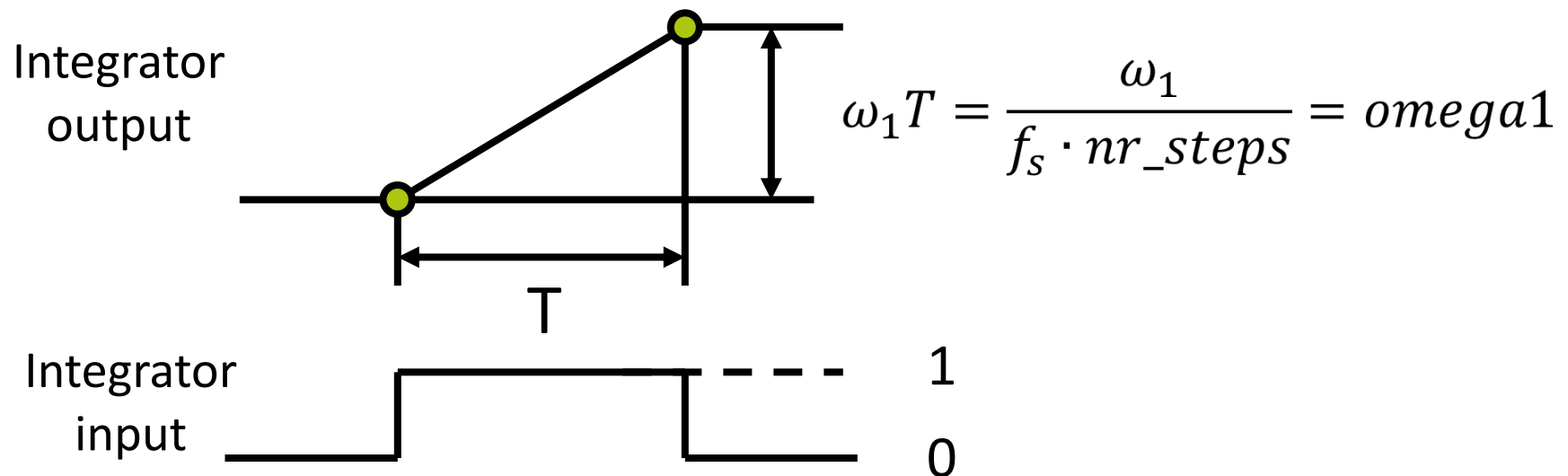
Code for Simulating a CT Modulator (2)

- How can we determine omega1 and omega2?
- Lets consider what happens when we apply a constant unit input to a CT integrator with a transfer function ω_1/s



Code for Simulating a CT Modulator (3)

- In the time domain, the integrator's output will be a linear ramp which will have grown by $\omega_1 T$ at the end of a calculation step where A is the amplitude of the input and T is the step duration



Exercise 2.7

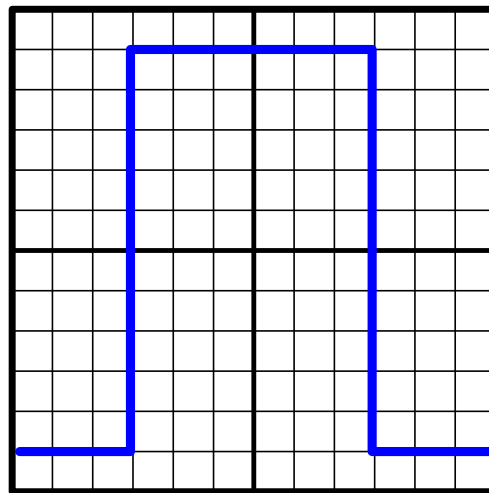
- Download (from Blackboard) and run the m-file: MOD2_CT.m
 - This file models a 2nd order CT sigma delta modulator with a feed-back topology.
 - Locate the two “for” loops simulating the modulator and the internal steps of the continuous time loop filter.
 - Locate the integrator unity gain frequency definitions, and try to trace their relation to the calculation step times.
- Modify the code to implement a feed-forward modulator
 - Size the modulator to reach specs of Assignment 3 (SQNR = 100 dB, SNR = 90 dB in BW=20 kHz)
 - Scale the coefficient to keep the integrator outputs < 0.5V

Exercise 2.7 - Solution

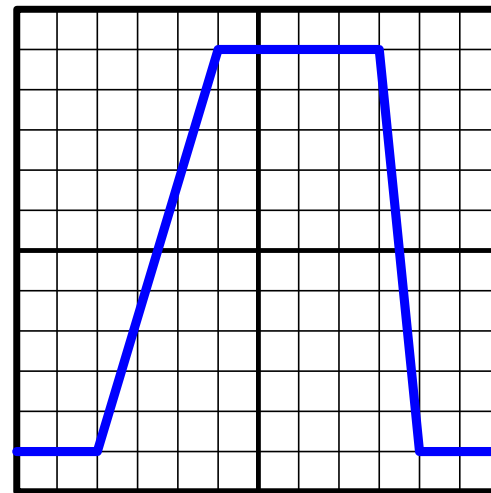
- Use the same OSR as discrete-time implementation and scale coefficient to get same shaping
 - $\text{OSR} = 250; \text{fb} = 20; \Rightarrow \text{fs} = 10 \text{ MHz}$
 - $\omega_1 = 2\pi\text{fs}/(\text{fs}\cdot\text{nr_steps})$
 - $\omega_1 = 2\pi\text{fs}/(\text{fs}\cdot\text{nr_steps})$
 - $c_1 = 4\pi$
- Add noise:
 - $\text{noise_rms} = \text{amplitude} \cdot \sqrt{\text{fs} \cdot \text{nr_steps} / 4 / \text{fb} / 10^{(90/10)}}$
- Scale:
 - $b_1 = 0.05;$
 - $b_2 = 0.055;$

Waveform asymmetry of DAC

- A practical single-bit DAC will in most cases output signals with asymmetric rise and fall times.
- This leads to a problem called: Intersymbol Interference (ISI) in continuous time sigma-delta modulators.
- ISI results in distortion, and more noise in the base band due to the intermodulation of harmonics with shaped Q-noise.



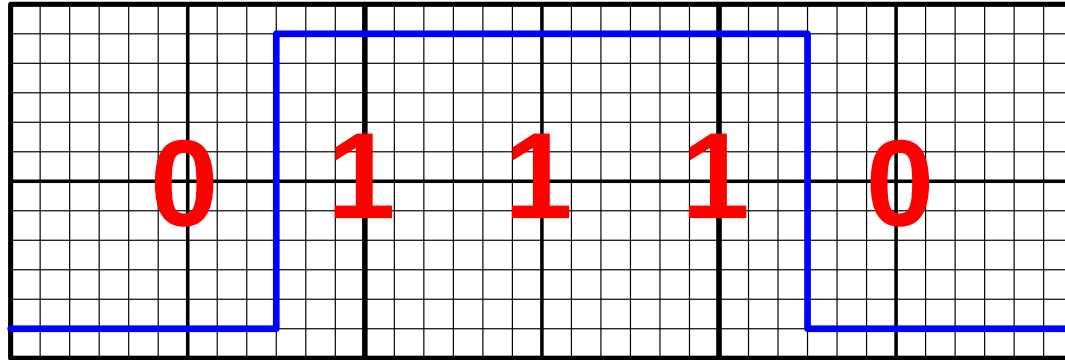
Ideal DAC signal



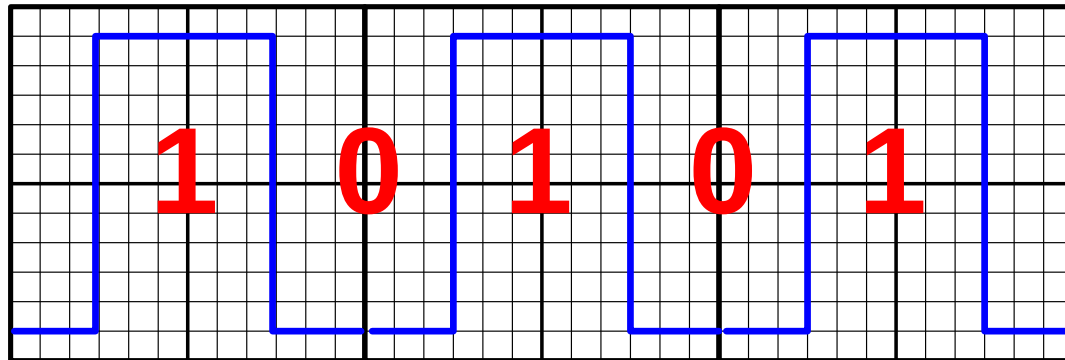
Asymmetrical DAC signal

No ISI

“01110” sequence



“10101” sequence

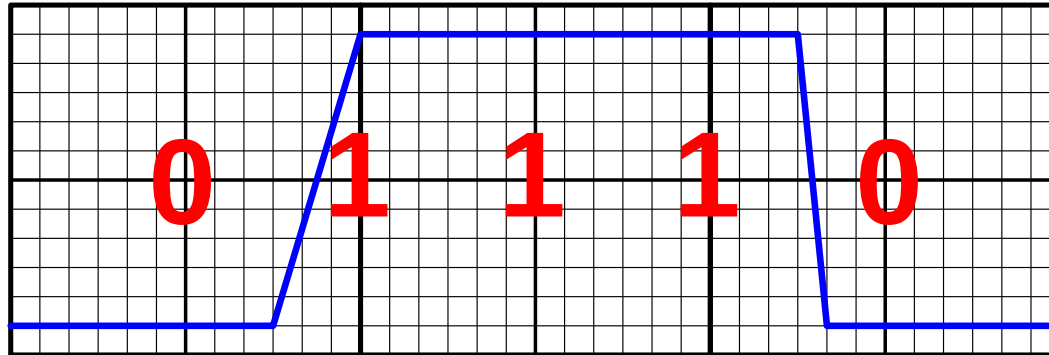


Ideal DAC signal

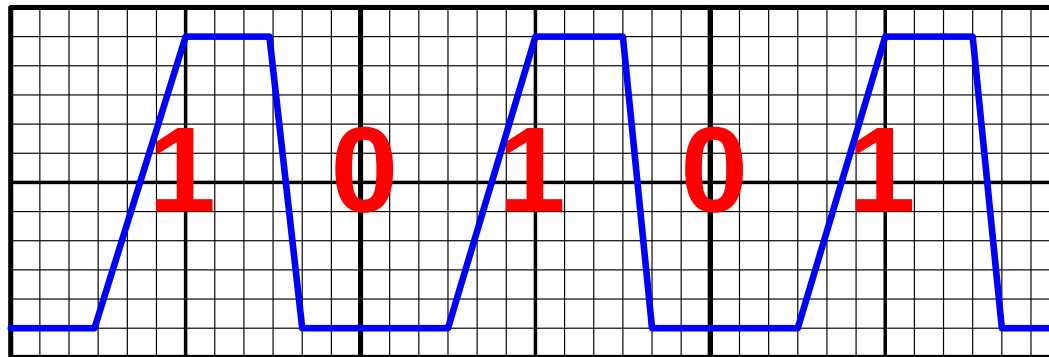
- Ideal DAC $\Rightarrow$ energy (area under the pulses) is the same for both “01110” and “10101” sequences.

ISI

“01110” sequence



“10101” sequence



Asymmetric DAC signal

- Non-ideal DAC $\Rightarrow$ energy of “01110” and “10101” sequences is **NOT** the same $\Rightarrow$ **data dependent** errors!

Exercise 2.8

- Download (from Blackboard) and run the m-file: MOD2_CT_NRZDAC.m
 - This m-file simulates a 2nd order continuous time **feed-forward** modulator with a sinusoidal input signal.
 - The modulator's single-bit DAC has **asymmetric** rise and fall times.
 - Go through the m-file, and observe how the CT DAC signal is defined and how the asymmetric rise and fall times are generated.
 - Run the code with asymmetrical DAC signals (default), and with ideal DAC signals (by removing the comments at proper place). Observe the output spectrum. What happens? Why?
 - Calculate the SNR and see how much resolution you lose due to the DAC non-ideality.